Δd is redundant, just d is fine

1.2 CHASE & COLLISION PROBLEMS

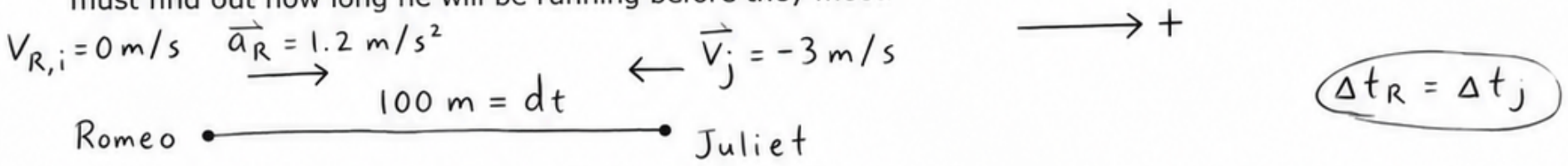
LEARNING GOALS

Students will solve problems involving two objects chasing each other or colliding

LOOKING FOR LINKS

With chase and collision problems, you won't have enough information to solve if you only consider one object; you have to consider both together and look for links between them. Consider the following examples:

EXAMPLE 1:
Romeo is sitting under a tree when he sees Juliet on the other side of the field (100. m away) running towards him at 3.0 m/s. He jumps up and accelerates towards her at 1.2 m/s^2. You must find out how long he will be running before they meet.



$\Delta t_R = \Delta t_j$

a) How do the times they are running for compare?
They are the same because they meet at the same time

b) They don't run the same distance, but if you add the distance each one travels, what will it be?
It will be 100 m because that's the total distance between them. $\Delta d_R + \Delta d_j = 100\ m$

We can write these symbolically as

$$\Delta t_R = \Delta t_J = \Delta t \quad (1)$$

$$\Delta d_R + \Delta d_J = 100\ m. \quad (2)$$

Next, we consider the equations of motion for both.

c) Juliet is moving with constant velocity so her distance can be calculated by the equation

$$\Delta d_J = |v_j| \Delta t \quad (3)$$

d) Romeo is accelerating. We know his initial velocity and acceleration and we're interested in time. Which equation of motion would be best?

$$\Delta d_R = V_{iR} \Delta t + \frac{1}{2} a_R \Delta t^2 \quad (4)$$

e) Now sub equations (3) and (4) into (2) and solve for time.

$\Delta d_R + \Delta d_j = 100\ m$

$\frac{1}{2} a_R \Delta t^2 + |v_j| \Delta t = 100\ m$

$\frac{3}{5} \Delta t^2 + 3 \Delta t = 100\ m$

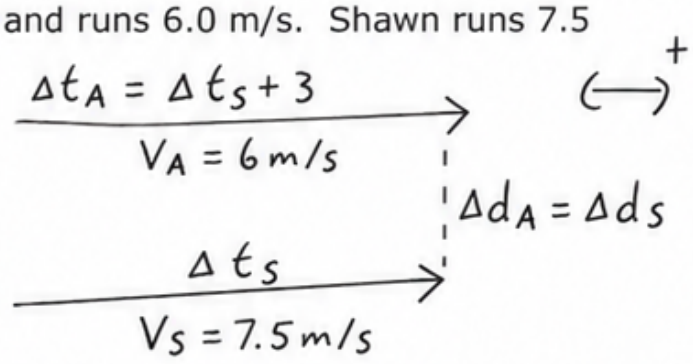
$\frac{3}{5} \Delta t^2 + 3 \Delta t - 100 = 0$

$\Delta t = 10.6\ s$ or $-15.6\ s$

10.6 s

EXAMPLE 2:

Shawn and Alvin are racing. Alvin gets a 3.0 s head start and runs 6.0 m/s. Shawn runs 7.5 m/s. How long until Shawn catches Alvin?



a) How do their distances compare?

They are the same

b) What is the connection between their times?

Alvin's time is Shawn's time plus 3 seconds

c) Both have constant velocity so their equations of motion are

$\Delta d_S = V_S \Delta t_S$ and $\Delta d_A = V_A (\Delta t_S + 3)$

d) Sub these into your equation from (a). BUT the TIMES are NOT equal so you'll have to sub in from (b) as well to solve.

$$\Delta d_A = \Delta d_S$$

$$V_A (\Delta t_S + 3) = V_S \Delta t_S$$

$$6 (\Delta t_S + 3) = 7.5 (\Delta t_S)$$

$$6\Delta t_S + 18 = 7.5 \Delta t_S$$

$$18 = 1.5 \Delta t_S$$

$$\Delta t_S = 12\ s$$

12 s

PRACTICE

Always make a sketch (except for the head start questions which are a little tricky to draw)

1. Two runners, moving at constant velocities, are at opposite ends of a straight 200 m track that runs north-south. Runner A starts out from the south end at 4.0 m/s [N]. Runner B starts out from the north end at an unknown velocity south. If both runners meet in 18 s, then what is the velocity of Runner B? (7.1 m/s)

2. Vancouver is about 300 km from Seattle. Two friends, one leaving from each city by bicycle, want to identify where to find a campsite between the two cities. One cyclist starts out from Vancouver at 25 km/h. His friend will start out 2.0 h later from Seattle at 32 km/h. How far from Vancouver do the friends meet? (160 km)

Chase + collisrons Practice

1)

V_B = ? m/s [S]

$\overline{V_A}$ = 4.0 m/s [N]

North (+)

$d_T = 200\,m$

$\Delta t = 18\,s$

South (-)

$d_A = V_A \Delta t \qquad d_B = V_B \Delta t$

$d_A + d_B = d_T$

$V_A \Delta t + V_B \Delta t = 200\,m$

$4\,m/s \times 18\,s + V_B \times 18\,s = 200\,m$

$72\,m + V_B \times 18\,s = 200\,m$

$V_B \times 18\,s = 128\,m$

$V_B = 128\,m / 18\,s$

$V_B = 7.1\,m/s\ [S]$ ← answer

2)

Vancouver (+)

$\vec{V_1}$ = -25 km/h

$\vec{V_2}$ = 32 km/h

$d_T = 300\,km$

Seattle (-)

$d_1 = |V_1| \Delta t_1$

$\Delta t_1 = \Delta t_2 + 2h \qquad d_1 = |V_1|(\Delta t_2 + 2h)$

$d_2 = V_2 \Delta t_2$

$d_2 + d_1 = d_T$

$V_2 \Delta t_2 + |V_1|(\Delta t_2 + 2h) = 300\,km$

$32 \Delta t_2 + 25(\Delta t_2 + 2h) = 300$

$32 \Delta t_2 + 25 \Delta t_2 + 50 = 300$

$57 \Delta t_2 + 50 = 300$

$57 \Delta t_2 = 250$

$\Delta t_2 = 4.4\,h$

$\therefore \Delta t_1 = 4.4\,h + 2h$

$\Delta t_1 = 6.4\,h$

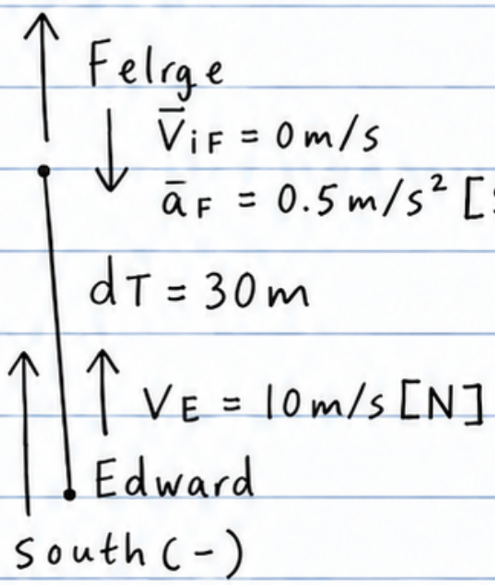
$d_1 = |V_1| \Delta t_1$

$d_1 = 25 \times 6.4$

$d_1 = 160\,km$ ← answer

3)

North (+)



$d_E = V_E \Delta t$

$d_F = V_{iF}\Delta t + \frac{1}{2} a_F \Delta t^2$ ($V_{iF}\Delta t = 0$)

$d_E + d_F = d_T$

$V_E \Delta t + \frac{1}{2} a_F \Delta t^2 = 30m$

$10\Delta t + \frac{1}{2}(0.5)\Delta t^2 = 30$

$10\Delta t + 0.25\Delta t^2 = 30$

$0.25\Delta t^2 + 10\Delta t - 30 = 0$

use quadratre formola

$$\Delta t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\Delta t = 2.8s$

~~$\Delta t = -43s$~~

4a) $\bar{\Delta} t = 360s$

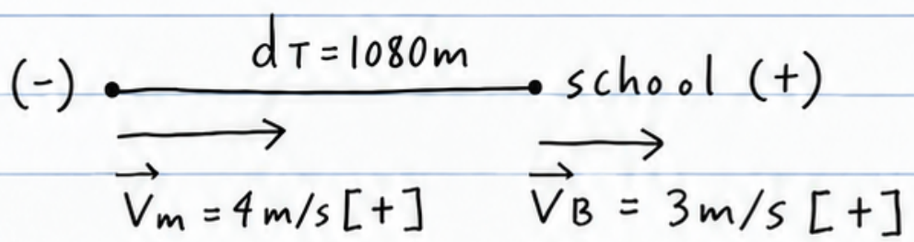
$d_B = ?$ → school (+)

$V_B \neq 15m/ss$

$V_B = 3m/s$

$d_B = \Delta t_B V_B$ $d_B = 360 \times 3$ $d_B = 1080m$

4b)



$\Delta t = ?$

$d_m = V_m \Delta t$

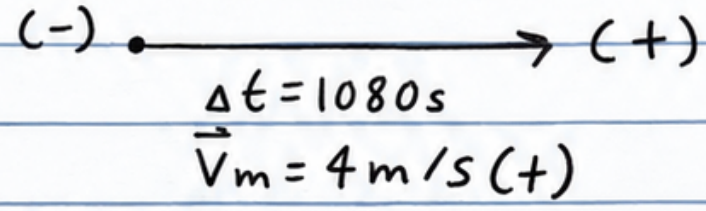
$d_B = V_B \Delta t + 1080m$

$d_m = d_B$

$V_m \Delta t = V_B \Delta t + 1080m$

$4\Delta t = 3\Delta t + 1080$

$\Delta t = 1080s$

4c) 

$d_m = V_m \Delta t$

$d_m = 4 \times 1080$

$d_m = 4320m$

Rest of the answers are on the digital version